FS51x. Changing perspectives: the science of optics in the visual arts

Aravi Samuel, Department of Physics

Art History is long intertwined with science and technology. The painter, to put what is seen by their eye onto flat canvas, has to grasp and solve technical problems. The brain will effortlessly create internal mental images of the three-dimensional surrounding world. But translating mental images into two-dimensional pictures – while retaining a sense of volume and depth, light and shadow, surface and texture – is hard. And so picture-making has long been perfused with optical science and technology – from early discoveries by Renaissance artists who labored to record the world with realism – to our cultural moment where every smartphone effortlessly captures and communicates visual experience. Here, we explore this story of art, picture-making, optics, and technology from the Renaissance to now. We ask how pioneering artists from van Eyck to Vermeer to modern photographers approached and solved problems in optical representation. How did artists across centuries capture visions of deeply-seen worlds in works that continue to captivate?

Figure 1: *The Crucifixion; the Redeemer with Angels; Saint Nicholas; Saint Clement* By Duccio di Buoninsegna (1311-1318). Note the physical realism in the fall and fold of every cloak over every human figure. Link to MFA

The story of Western art is a story of evolving style across different eras, like the emerging realism in the Italian Renaissance (Figs. 1, 2) to the near-photographic realism of the Dutch Republic (Figs. 3, 4, 5) to contemporary art and photography (Fig. 9). Ambitious artists pioneered new schools of art when they succeeded in inventing new styles of visual representation. To do this, many artists helped themselves to inspiration from contemporary scientists. Innovations in the science and technology of image-making could lead to stylistic shifts in art history.

Figure 2: *The Annunciation* by Piermatteo d'Amelia (1487) Note the mathematical precision of a central vanishing point and the illusion of three-dimensional space. Link to Gardner Museum

Artists and scientists have something in common. Both want to have original ideas about their world and to communicate these ideas to others. But whereas science strives for objective fact and unambiguous communication, art invites subjectivity. Every artist and beholder has a different point of view. The meaning of any artwork is not fixed or finite, but flexible.

But art is also not arbitrary. Every artwork is made and seen within objective realities: in the technical act of making marks on surface, in the optical physics of light and shadow, and in the neurobiology of the eye and brain. Understanding these objective realities opens windows into works of art.

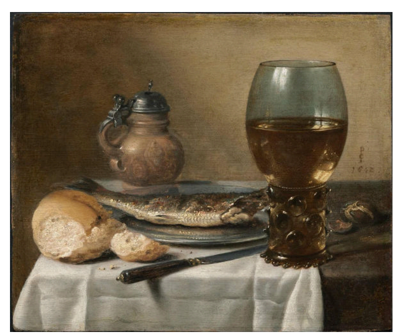

Figure 3: *Still Life with Stoneware Jug, Wine Glass, Herring, and Bread* Pieter Claesz (1642). Note the optically precise rendering of every reflection and refraction. Link to MFA

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Figure 4: *View of Delft* by Vermeer, 1660-1661. Link to Mauritshuis.

CONSIDER VERMEER. Johannes Vermeer (1632-1675), the beloved painter of *The Girl with a Pearl Earring*, has also been called the most 'photographic' painter of the Dutch Republic. Optical realism saturates his *The View of Delft* and *The Music Lesson*, capturing every play of light and shadow with subtlety and precision. Look carefully throughout the *Music Lesson*. Doubled shadows are cast by the mirror frame onto the wall, just what happens when light projects from multiple directions from multiple windows. The surfaces of the curved pitcher capture glinting images of sunlight through the same windows that smoothly modulate into shadows. Vermeer pays close attention to linear perspective – every set of parallel line on the tiled floor, walls, and ceiling converge to common vanishing points – the mathematical result when geometrically flattening a three-dimensional scene onto a two-dimensional plane. The painted mirror on the wall reflects what is both in front and behind the picture plane, catching a glimpse of the legs of the painter's easel.

Figure 5: *The Music Lesson* by Vermeer c.1662 - 1665.

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VERMEER may have been artistically inspired toward 'opticality' by contemporary Dutch science. Vermeer worked in the 17^{th} century Dutch Republic. In the same era, Christiaan Huygens (1629-1695) developed the astronomical telescopes that discovered Titan, Saturn's largest moon; and Antonie van Leeuwenhoek (1632-1723) developed the microscope that discovered microbial life. Vermeer and Leeuwenhoek were born in the same year and the small town of Delft. Perhaps Leeuwenhoek was the model for the young man in Vermeer's *Astronomer* and *Geographer* (Fig. 6, 8). In both artworks, Vermeer portrays scientists who share his commitment to faithful visual recording of the natural world.

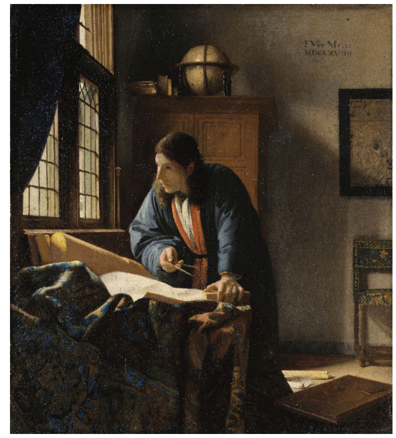

Figure 6: *The Geographer* by Vermeer and its Link at Google Arts and Culture.

Figure 7: *Portrait of Antonie van Leeuwenhoek* by Jan Verkolje 1680 - 1686

Figure 8: *The Astronomer* by Vermeer and its Link at Google Arts and Culture.

Figure 9: *Ready-made with Skull and Mirrors* by David Hockney, 2018. A 'photographic drawing' that revels in optical reflection.

David Hockney, the contemporary artist, is inspired by opticality in visual art. His *Ready-made with Skull and Mirrors* (Fig. 9) is a 'photographic drawing' that reflects Hockney's interest in optics and perspective. He depicts a complex ensemble of objects and multiply-reflecting mirrors in an invented, but optically convincing, three-dimensional space. Hockney is a classically-trained artist who can do much with just pencil or paintbrush. But here, Hockney used modern tools to plot and construct the scene.

Hockney wondered about scientific and optical devices in paintings by Vermeer and others. He speculated that the very same devices might have been useful, not just as props in those paintings, but as *tools* to help *make* pictures. His 'Hockney Thesis' is that artists like Vermeer were not just *inspired* by optics but *used* optical tools like lenses and mirrors to paint and draw. He published this controversial idea in *Secret Knowledge* (Fig. 10). The book cover shows Hockney drawing with a *camera lucida*, a drawing device based on a prism that was invented in the 1800s.

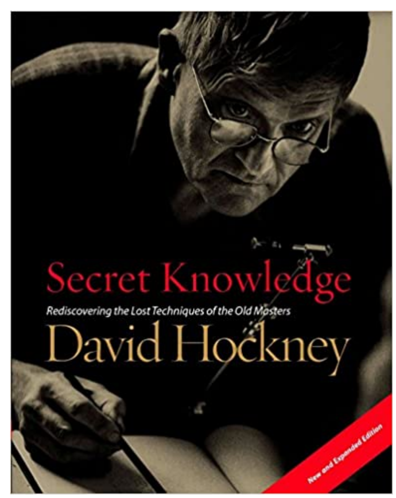

Figure 10: David Hockney using a camera lucida on the cover of his book.

SEEING, KNOWING, AND COMMUNICATING is the work of both artists and scientists. This course is a journey into how artists responded to their visual world across Western Art History from perspectives of science and technology. This includes:

- **Making Art**. Many students might not be artists. We will learn about painting and drawing from artists who use optical tools to make art, including Ethan Murrow, a Boston-based muralist (Fig. 11).
- **Experiments**. Appreciating the interplay between science and art requires forms of intuition best gained by doing experiments. For example, we will learn about mirrors and lenses, reflection and refraction by *seeing* how they work, not by solving equations.
- **Field Trips**. We will visit nearby museums, the Harvard Art Museums, the Gardner Museum, and the MFA. We will study original artworks, and think about their connections to contemporary science and technology.

WE WILL USUALLY MEET in Harvard Art Museums 0600. Except during field trips when we travel to museums. We will sometimes use HAM's Art Study Center to look closely at works in Harvard's collection. We will sometimes use HAM's M-Lab, its maker space, to make art. We meet Thursdays from 12:45–2:15 PM.

READING AND WRITING ASSIGNMENTS will be assigned each week. Each week, students will read one or two short papers or book chapters. Each week, students will write a short essay (2-3 pages) that responds to readings and classroom discussions.

WE WILL BE SUCCESSFUL if understanding science and optics deepens our appreciation and interest in pictures and paintings, pushes us to look more carefully at the world around us, and encourages our own picture-making.

Contents

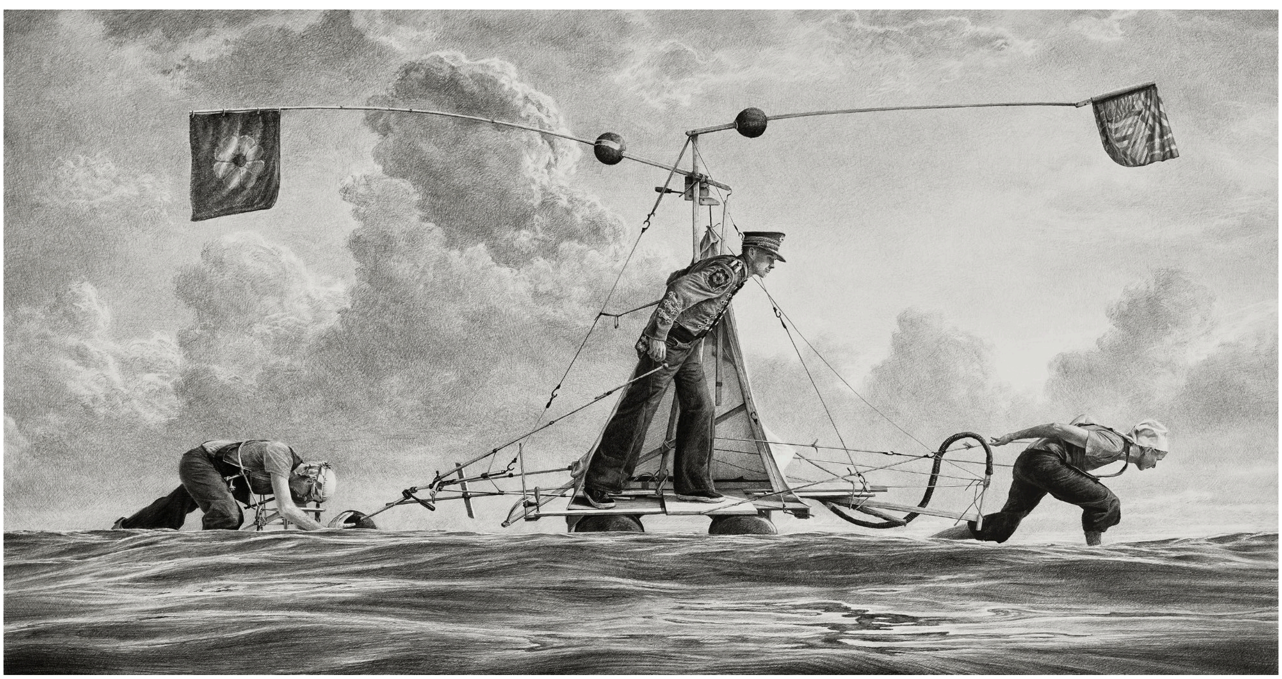

Figure 11: *For All Intents and Purposes* by Ethan Murrow. Graphite on Paper, 52" x 100".

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Dramatis Personae

Aravi Samuel '93 studied physics and biophysics at Harvard as an undergraduate and graduate student with Prof. Howard C. Berg. Berg, a microscopist of microorganisms, introduced Aravi to Leeuwenhoek, Vermeer's contemporary. Aravi's email: samuel@physics.harvard.edu.

Claire Swadling '26 studies physics and has studied Studio Art. Contact her about course material, scheduling and transportation, and other questions. Claire's email: cswadling@college.harvard.edu.

Frances Yee '27 took this class! Frances is an accomplished artist. Contact her about course material and other questions. Frances's email: francesyee@college.harvard.edu.

Susan Schmidt, from the Harvard Science Center, supports our physics experiments and demonstrations.

Ethan Murrow is Professor at The School of the MFA. He focuses on historical narratives, large scale wall drawings and murals. Ethan's website is here.

Penley Knipe is Philip and Lynn Straus Senior Conservator at the Harvard Art Museums. Penley is an expert in the history and techniques of American portrait silhouettes.

Joachim Homann is Maida and George Abrams Curator of Drawings at the Harvard Art Museums. Joachim acquired many of the original drawings made with optical tools for the Harvard museums.

Chris Atkins is Van Otterloo-Weatherbie Director of the Center for Netherlandish Art (CNA) at the MFA, a center for scholarship on Dutch and Flemish art.

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Figure 12: *School Class with a Sleeping Schoolmaster* by Jan Steen, 1672

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Assignments

Deep looking, Due February 12

Professor Jennifer Roberts developed the exercise of asking students to look deeply at a painting for 3 hours. Read her worksheet, and do a shorter version of her exercise.

Choose a richly detailed picture in the Harvard Art Museums. Look at it for 1 minute. Then turn away and write down what you saw. Then look at it for a full 15 minutes. Turn away and describe what more you were able to see.

Color, Due February 19

Find and describe a painting in the Harvard Art Museum where color is used more than decoratively, where the artist makes careful and deliberate use of color to compose the image.

Shadow, Due February 26

Use your phone cameras to capture a surprising or intriguing shadow effect, whether something you stumble upon or something you arrange. Use words to describe the optics of what is happening in your photograph.

Perspective, Due March 5

Find and describe a painting in the Harvard Art Museum where perspective is used rigorously or intuitively, and identify where the artist makes choices in their depiction of three-dimensional space.

Seeing and believing

Thursday, 29 January 2026, 12:45 - 2:15 PM EST.

Artists and audiences translate between mental and physical pictures when making or viewing art. An artist's views about the nature of vision can shed light on their artistic approach. Such views have changed alongside scientific progress in our understanding of the physical and biological nature of light and vision. Even now, our understanding of human vision is changing in response to recent advances in artificial intelligence. We can now build machines that see and recognize objects as well as humans do. How the human brain creates mental pictures from visual images remains largely unexplained, but we have gained new clues from AI systems that became markedly better at parsing and labeling visual images after incorporating brain-inspired circuit architectures.

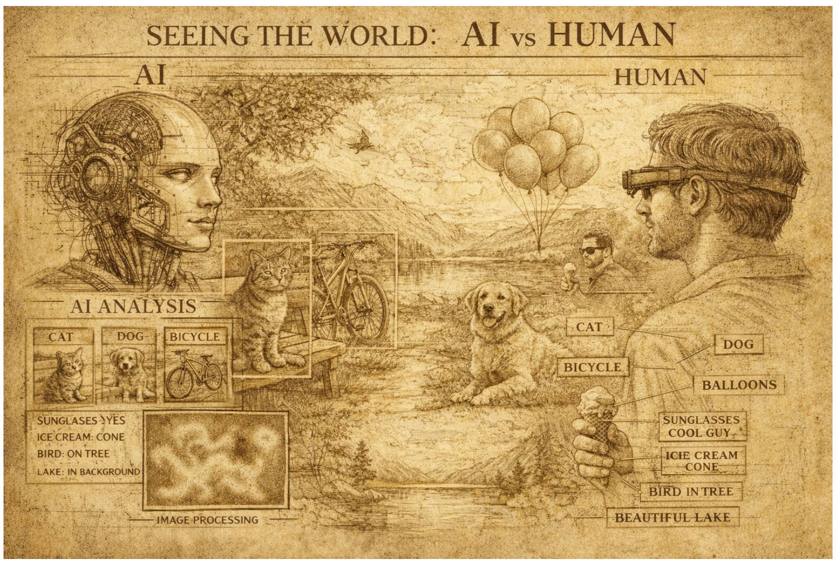
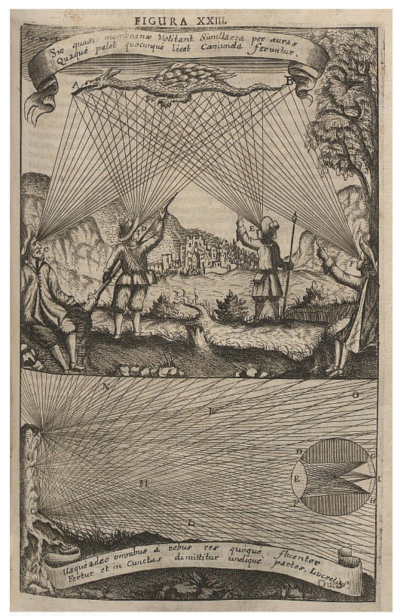


Figure 13: AI-driven image recognition systems can now segment and label objects within visual images using biologically inspired neural-network architectures.



Quelle: Deutsche Fotothek

Figure 14: Illustration from *System der visuellen Wahrnehmung beim Menschen* (1687), depicting emission theory.

Artificial intelligence is opening a new chapter in the history of making and viewing art. But let us turn back a few pages. Before Newton, Maxwell, and Einstein recast light as waves and photons, vision was explained through speculative philosophical models, not empirical physical laws. Aristotle (384-322 BC) and Alhazen (c. 965-1040) favored **intromission** theory, in which the eye receives light of purely external origin. Galen (c. 129-216) favored **extramission**, in which the eye emits outward rays that see objects by 'touching' them. Roger Bacon (1219-1292) compromised with a hybrid theory in which the eye emits rays that interact with objects and then return to the eye with visual information.

From a modern perspective, it is obvious that intromission is the only tenable theory of human vision. But until Kepler (1571-1630) discovered the optics of image formation by the eye — decisively debunking extramission and Bacon's compromise — alternate philosophies held sway. These philosophies also found their way into art.

In the Italian Renaissance, one of the most frequently painted biblical scenes was the announcement by the Angel Gabriel to the Virgin Mary that she would bear Christ. These *Annunciation* paintings became visual windows into pre-modern theories about light and vision. As described in the Book of Luke (1:26-38):

> To a virgin pledged to be married to a man named Joseph, a descendant of David. The virgin's name was Mary. The angel went to her and said, "Greetings, you who are highly favored! The Lord is with you."
>
> Mary was greatly troubled at his words and wondered what kind of greeting this might be. But the angel said to her, "Do not be afraid, Mary; you have found favor with God. You will conceive and give birth to a son, and you are to call him Jesus. He will be great and will be called the Son of the Most High. The Lord God will give him the throne of his father David, and he will reign over Jacob's descendants forever; his kingdom will never end."
>
> "How will this be," Mary asked the angel, "since I am a virgin?"
>
> The angel answered, "The Holy Spirit will come on you, and the power of the Most High will overshadow you. So the holy one to be born will be called the Son of God. Even Elizabeth your relative is going to have a child in her old age, and she who was said to be unable to conceive is in her sixth month. For no word from God will ever fail."
>
> "I am the Lord's servant," Mary answered. "May your word to me be fulfilled." Then the angel left her.

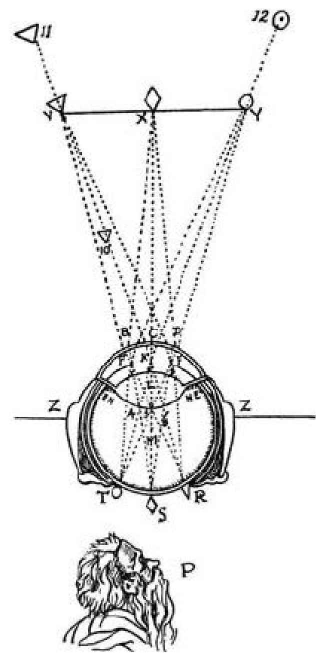


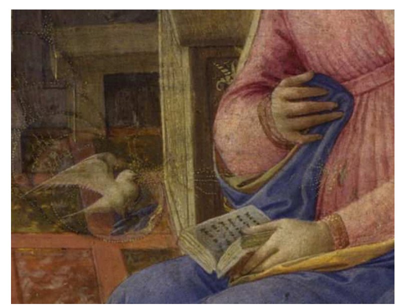
Figure 15: Illustration of the theory of the retinal image from *La Dioptrique* by Descartes. In 1605, Kepler discovered that light entering the eye is focused by the lens, forming a real, inverted image on the retina, in the same way that a camera focuses an image onto its photosensitive surface.

A key motif of *Annunciation* paintings is the Word of God as the agent of immaculate conception. The Word of God was typically cast as a beam of light descending on Mary, often conveying a dove signifying the Holy Spirit. Translating the audible Word of God into a visible beam of light was an ingenious artistic invention. The immaculate conception was thus conceptualized as an act of seeing, mediated by an intangible beam of extramission. A weighty theological problem — conceiving Christ without the messiness of physical contact — was thus lightly solved by clever artists.

In Carlo Crivelli's *Annunciation, with Saint Emidius* (1486), a beam of golden light shoots from an opening in the sky, passes through a window, and reaches the Virgin Mary. The dove rides along a ray of extramission from God to Mary.

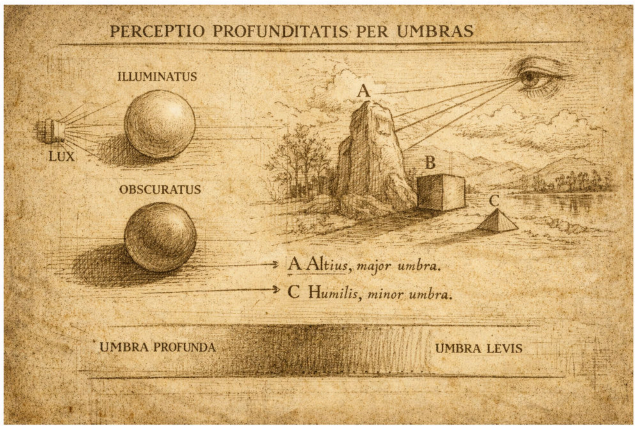
Figure 16: *The Annunciation* by Carlo Crivelli, 1486. Link to National Gallery

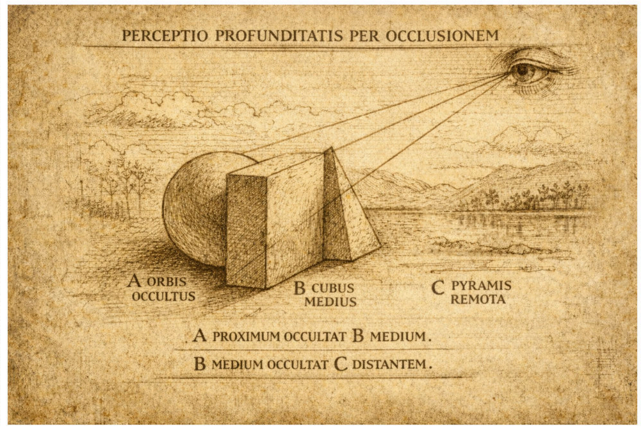
Figure 17: *The Annunciation*by Filippo Lippi, c.1449-1459, in the National Gallery, London is painted as a lunette that was likely placed above a door or bed. Link to National Gallery



Leo Steinberg (1920-2011), an American art critic and historian, noticed a tiny detail in one painting that made it unique among many thousands of *Annunciation* paintings. Fra Filippo Lippi (1406-1469) was an important Florentine painter in the Italian Renaissance who produced many *Annunciations* in his lifetime. His London *Annunciation* may be his most mature version. In an unusual departure from convention, the dove representing the Holy Spirit hovers close to Mary. On closer inspection, Steinberg identified delicate golden rays of extramission extending from the dove toward a slit in Mary's tunic. Uniquely in this *Annunciation*, Mary's womb also returns the golden rays. Lippi may have become aware of the hybrid theory proposed by Roger Bacon (1219-1292), the English friar who sought to reconcile Catholic faith and natural philosophy. Steinberg's research led him to St. Antoninus (1389-1459), a Dominican friar who preached in Florence about Bacon's ideas at the time Lippi painted this work. Having encountered Bacon's ideas, Lippi may have translated them into paint. Believing, in this case, is seeing.

Kepler's discovery that the lens of the human eye focuses images onto the retina leads to new insights. First, it explains our natural comfort with two-dimensional pictures. From cave paintings to iPhone screens, almost all human visual communication is done with flat images. We effortlessly use two-dimensional images because all human seeing is fundamentally flattened. Flat paintings and drawings vastly outnumber solid sculptures. But even when looking at a sculpture, we only see its flattened retinal image. Binocular vision with two eyes helps us see depth, but only at short distances. Most depth perception is not binocular, but by using visual clues like shadowing or the occlusion of objects in front of one another.





The flatness of the retina in a biological eye and the flatness of the light-sensitive chip/film in a photographic camera is a coincidence that can lead to misconceptions. When artificially-intelligent machines 'see' an image, they simultaneously gather information from every pixel. We feel like we see with continuous high-definition. But this is an illusion. Unlike flat light-sensitive machines, our retinas are not homogeneous. Each human retina has only one tiny region that sees with high-spatial resolution called the fovea. The fovea is densely packed with small cone photoreceptor cells. We see color using three types of cone cells, each with peak sensitivity to different wavelengths of light. Another photoreceptor, called the rod cell, is only able to dim lights during night vision. During the day, rod cells are inactive.

Inside the fovea, the small size and dense packing of cone cells confers high spatial resolution. But outside the fovea, >1° of angular seeing, cones decrease in number and enlarge in size, lowering spatial resolution. This is revealed by self-experiment. Focus your eye on a single word on this page, reading while forcing your eyes to remain still. How many surrounding words can you read? The answer is very few. Non-foveal peripheral vision is too low to read even at short distances.

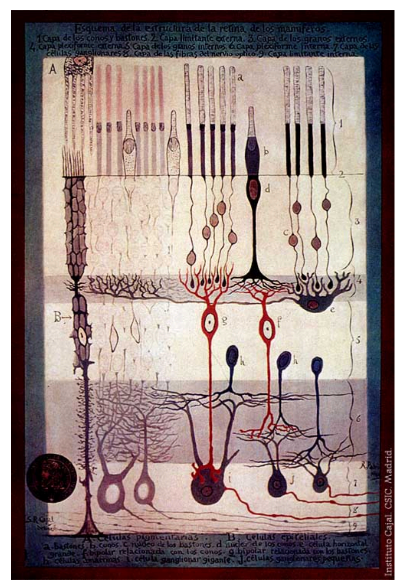


Figure 18: *Structure of the Mammalian Retina seen in cross-section* c.1900 By Santiago Ramon y Cajal. a. Rods. b. Cones. c. Rod nucleus. d. Cone Nucleus. e. Large horizontal cell f. Cone-associated bipolar cell. g. Rod-associated bipolar cell. h. Amacrine cells. i. Giant ganglion cell. j. Small ganglion cells.

The human visual system is an inference machine. Unlike AI-vision which is spatially uniform and simultaneous, human vision is sequential, patchwork, and constructive. Our feeling that our whole visual field is being seen at high-resolution is an illusion of high-level brain processing. Most of the time, our eyes are flitting across the visual field, making hundreds of saccades every minute. Each saccade grabs a small piece of the visual field at high-resolution. With unconscious visual processing, the brain seamlessly assembles these pieces into a visual whole, filling in blanks using memories of what things are supposed to look like. When visual information is ambiguous, the brain has to guess. Thus, human vision is prey to illusion, at every disconnect between available visual information and physical reality.

Our brains see by selecting the most plausible interpretation of available information. Ernst Gombrich explored this idea in his landmark *Art and Illusion*. The fundamental issue is illustrated by the duck/rabbit illusion. Duck or rabbit? The available visual information can lead to a mental picture of either animal. Perception can wobble between duck or rabbit. Both animals can be seen, but not simultaneously.

The duck/rabbit illusion works because the drawing provides partial visual information for each animal. Neither is drawn with photographic precision. The ambiguous drawing has enough duck-like and rabbit-like features that allow the brain to see either animal. Similarly, the Rubin vase uses a silhouette to reduce visual information, creating an ambiguity between two faces and goblet. Faced with partial information, the brain has to choose what it perceives. Gombrich realized that every artwork, every man-made drawing or painting, is an illusion based on limited information.

Figure 19: The duck-rabbit illusion first appeared in *Fliegende Blätter*, a German magazine. It was later used by Ernst Gombrich in *Art and Illusion* and Ludwig Wittgenstein in *Philosophical Investigations*.

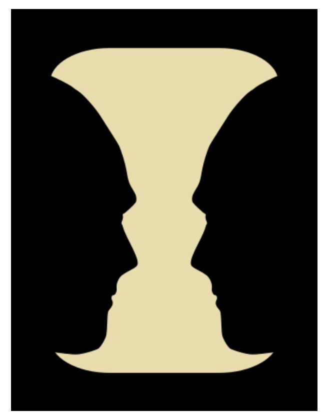


Figure 20: The Rubin vase is an example of ambiguous or bi-stable two-dimensional figure developed around 1915 by the Danish psychologist Edgar Rubin.

Billy Pappas (1984-) spent 8 years maximizing the visual information in one pencil drawing. Throughout the process, Pappas hung his arms from slings to keep them still. One hand held a magnifying glass. The other held a freshly sharpened pencil. After applying 9,000,000 pencil marks to one sheet of paper (12.8"x16.3"), he completed his *Marilyn Monroe*. At the end, the picture remains an illusion. A pencil can't draw blonde hair. The brain fills in the perception of Marilyn's hair where the pencil does not make a mark.

Figure 21: *Detail from Marilyn Monroe* by Billy Pappas, 2003.

Billy Pappas, *Marilyn Monroe*, 2003. Graphite on paper, 12.79 x 16.34 in. Courtesy of the artist. Digital image provided courtesy of William A. Christens-Barry, Chief Scientist, Equipoise Imaging, LLC.

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WE SEE WHAT WE BELIEVE SHOULD BE SEEN. Long before Billy Pappas, Jan van Eyck (c. 1390 - 1441) made a pioneering leap toward photographic realism in Early Netherlandish Painting. Giorgio Vasari (1511 - 1574) started the study of Art History with his *Lives of the Most Excellent Painters, Sculptors, and Architects.* Vasari claims that Jan van Eyck invented oil painting. Before van Eyck, pigment was usually mixed into egg yolk (*tempera*) a permanent, fast-drying, water-soluble, and long-lasting medium. With tempera, the artist has to hurry, having to get things right before the hardening of the permanent and fast-drying paint. Because linseed oil mixed with pigment takes *weeks* to dry, an oil painter has the luxury of time to refine every detail. The invention of oil painting coincides with a sharp rise in photographic realism in the Early Renaissance.

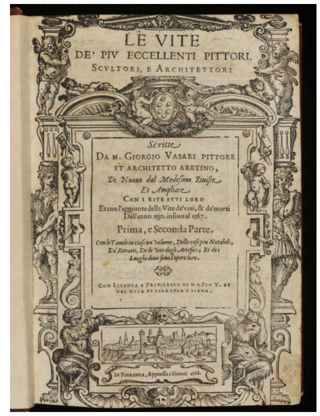
LE VITE
DE' PIV ECCELLENTI PITTORI,
SCVLTORI, E ARCHITETTORI
Scritte
DA M. GIORGIO VASARI PITTORE
ET ARCHITETTO ARETINO,
Di Nuouo dal Medesimo Riuiste
Et Ampliate
CON I RITRATTI LORO
Et con l'aggiunta delle Vite de' viui, & de' morti
Dall'anno 1550. insino al 1567.
Prima, e Seconda Parte.
Con le Tauole in ciascun volume, Delle cose piu Notabili,
De' Ritratti, Delle vite degli Artefici, Et dei
Luoghi doue sono l'opere loro.
CON LICENZA E PRIVILEGIO DI N. S. PIO V. ET
DEL DVCA DI FIORENZA E SIENA.
IN FIORENZA, Appresso i Giunti 1568.

Figure 22: Title page of the first and second part of the 1568 edition of Vasari's *Lives*.

JAN VAN EYCK probably didn't invent oil painting, but he did pioneer its new possibilities. One monumental achievement was *The Ghent Altarpiece* polyptych, designed and begun by his brother Hubert in the 1420s and finished by Jan in 1432. In the *Milites Christi* panel, van Eyck paints single hairs of the horse's mane and drops of spittle from its mouth, with near microscopic attention (see penny at scale).

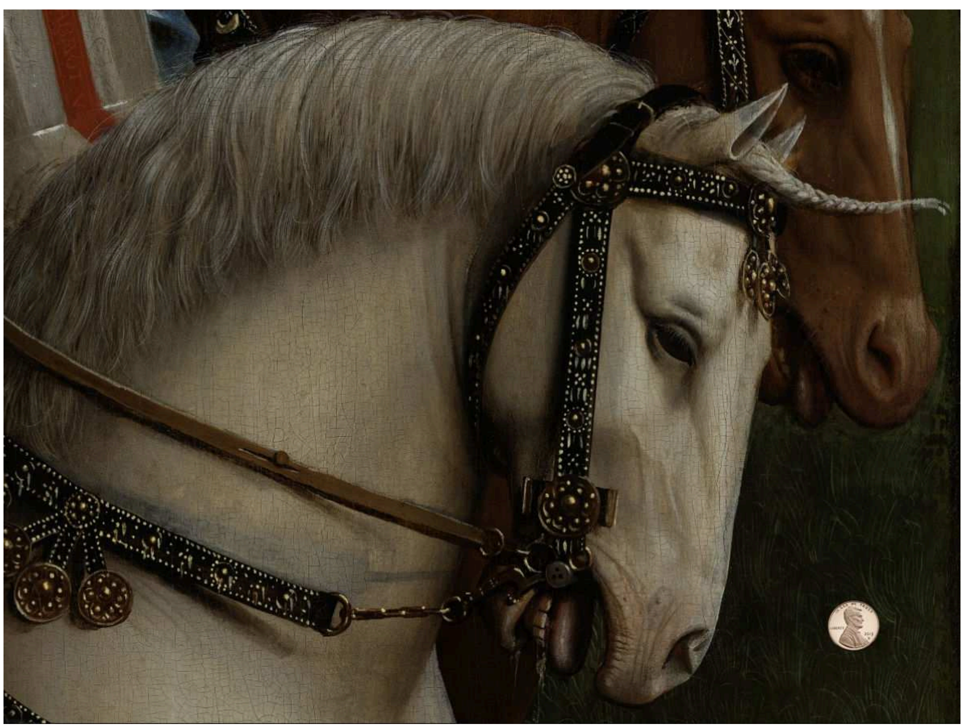

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Figure 23: *Milites Christe*

Figure 24: *The Ghent Altarpiece* by Hubert and Jan van Eyck. Closer to van Eyck.

THE EUROPEAN RENAISSANCE recovered scientific knowledge from Greek and Roman antiquity that had been preserved in the East. During the European Gothic Age – so named to blame the Goths that felled Rome – the Arab world enjoyed its Islamic Golden Age. The Egyptian Alhazen (or Ibn al-Haytham) (965-1040) was its leading mathematician, physicist, and astronomer. Alhazen's work was not widely known until 1200 when his *Kitāb al-Manāẓir* was translated into Latin as *De aspectibus* or *Perspectiva* (or the *Book of Optics* in English). Herein, Alhazen established principles of reflection and refraction with differently-shaped surfaces and transparent objects. European scientists including Newton, Kepler, Copernicus, and Huygens cited Alhazen's work, making him the true father of modern optics.

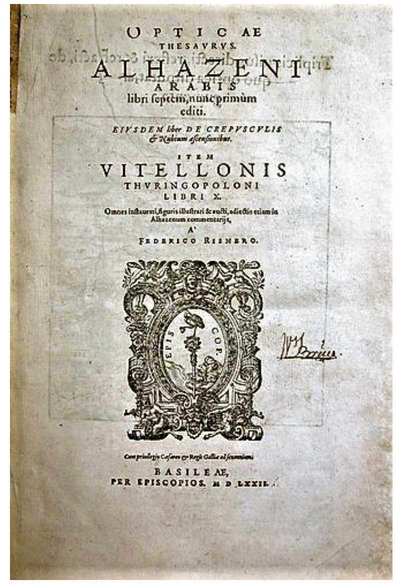
OPTICAE THESAVRVS.
ALHAZENI ARABIS
libri septem, nunc primum editi.
EIVSDEM liber DE CREPVSCVLIS & Nubium ascensionibus.
ITEM
VITELLONIS THVRINGOPOLONI LIBRI X.
A FEDERICO RISNERO.
BASILEAE,
PER EPISCOPIOS. M D LXXII.

Figure 25: Cover page for Alhazen's *Book of Optics* in the print edition from 1572.

Alhazen's work would have been available to the well-educated van Eyck. Was van Eyck himself a careful scholar of subtle optical effects? Or did awareness of Alhazen prepare van Eyck to notice and incorporate optical effects into his paintings?

Consider the crystal staff wielded by the Deity in the Ghent Altarpiece. The light that falls on and through the crystal shaft is refracted as a bright line along the hand, just like a cylindrical lens. Consider the glass and pearl beads hanging from his gold brocade. Specular reflections appear on the front surfaces of these curved beads, soft and diffuse on the pearl and sharp and pointed on the glass beads. Look behind the glass beads. Here, Van Eyck paints both their shadows, and bright focal points. Glass beads are magnifying glasses that concentrate incoming beams of light.

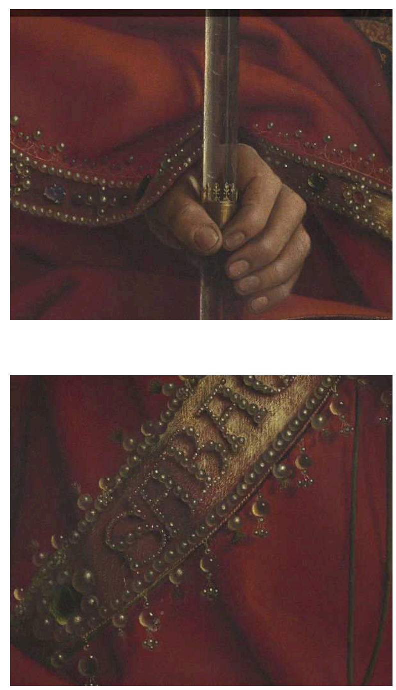

Figure 26: Optical details from the *Ghent Altarpiece*

Figure 27: Illustration of specular reflection and focused light collected by the spherical lens.

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THE ARNOLFINI PORTRAIT by van Eyck is another display of optics. On the back wall, van Eyck paints a mirror that reflects the scene in front of us and the scene in front of the painting. Look carefully — the artist includes himself in the reflection. The spherical mirror is both a virtuoso display of painterly skill and the artist's deep interest in optical effects. Van Eyck's spherical mirror is so carefully painted that its curved reflections are mathematically consistent with the geometry of the room. Convex mirrors with mathematically coherent reflections appear in many Early Netherlandish paintings, including a purported copy of an original painting by van Eyck (Fig. 29).

Figure 28: *The Arnolfini Portrait* by Jan van Eyck (1390-1441) in the National Gallery of London, and Link to all of van Eyck's Paintings.

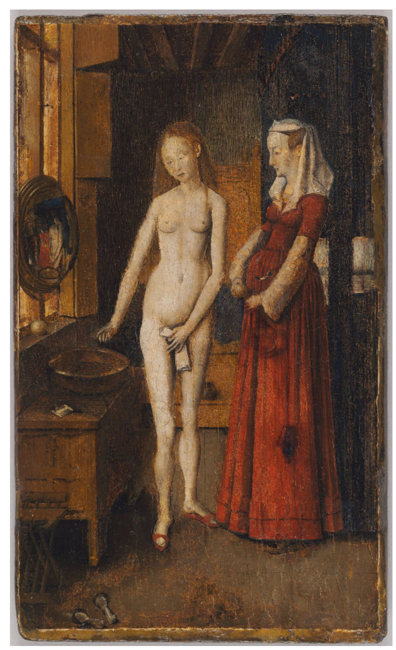

Figure 29: A contemporary of Jan van Eyck claimed that the artist painted a number of pictures like this one. Although these paintings are now lost, this panel, dating to the early sixteenth century, is believed to be a copy after one such work. Link to painting in the Harvard Art Museums.

Updated: February 19, 2026

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Interaction of Color

Thursday, 5 February 2026, 12:45 - 2:15 PM EST.
Harvard Art Museums 0600

Figure 30: *A painter in his studio* Rembrandt Harmensz van Rijn (Anonymous pupil of) 1600-1700. Link to painting.

We see with four different types of photoreceptor cell. In dim light, we see with rod cells. In bright-light, we see with three cone cells that are tuned to different wavelengths of visible light. This is why rod vision is monochromatic and cone vision is trichromatic. Because the brain uses the activity of three different cone cells to decide what color it sees, any color that we see can be mixed from three different hues, such as red/green/blue or cyan/magenta/yellow.

Trichromatic vision is helpful to the painter, letting them create any visible color by mixing only a small number of pigments. The artist in Rembrandt's studio only appears to use seven pigments on his palette (Fig. 30). Leonardo da Vinci wrote that he only needed four different pigments – yellow, blue, green, and red – to create all colors. Yellow and blue make green. So da Vinci could have gotten away with three pigments.

Why do we see the specific wavelengths of visible light? Visible light happens to be inside a bandwidth of solar radiation that penetrates Earth's atmosphere. Daylight is evenly distributed across the wavelengths that we see, from 400 nm (blue) to 700 nm (red).

Newton discovered that white daylight could be separated into the spectrum of different colors using a prism (Fig. 31).

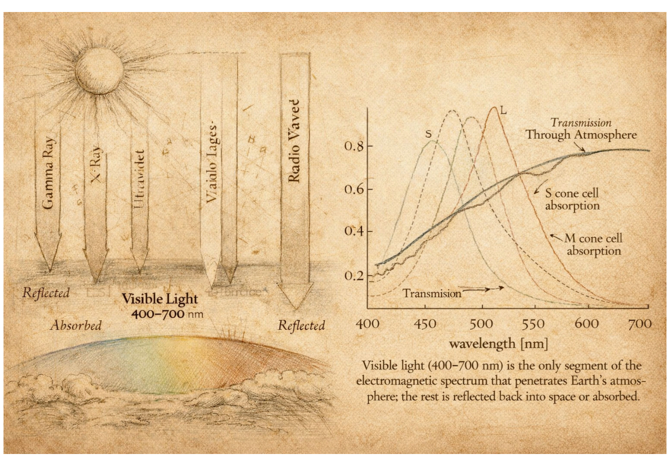

Figure 31: Most wavelengths of light are blocked by the atmosphere, except the visible bandwidth and X-rays. The latter is useful for X-ray telescopes. The spectrum of visible light is defined by a photon arrival rate function that is nearly constant over the range of visible light. From Nelson (2017)

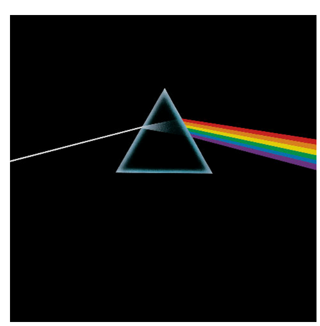

Figure 32: *Dark Side of the Moon*, Pink Floyd.

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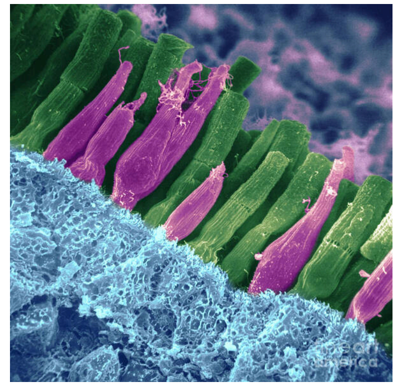

Figure 33: Art Print of pseudocolored electron micrograph of the retina, with green rod cells, red cone cells. From *www.pixels.com*.

CONE CELLS are most densely packed in the fovea, the center of the field of view. The diameter of each cone cell is roughly on the order of the wavelength of visible light – a biological limit to spatial resolution. When you try to see spatial details that are smaller than your spatial resolution, the details will blur.

OUR SCIENTIFIC UNDERSTANDING OF THE LIMITS OF SPATIAL RESOLUTION came with the anatomical studies of Santiago Ramón y Cajal (1852-1934). Around the same time, George Seurat (1859-1891) and other pointillists were doing their own experiments involving the spatial resolution of color vision. Instead of brushstrokes, Seurat painted with small colored lights, individual dots and dashes of different color.

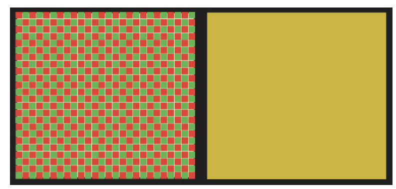

Figure 34: **Color illusion.** When viewed up close, the left box is seen to consist of small red and green squares. The image of each square on the back of the retina is larger than the size of photoreceptor cells. When viewed from afar, each photoreceptor receives light from both red and green squares, whose spectra merge. The perceived color becomes closer to the color in the right box than red or green. Red + Green $\sim$ Yellow.

VIEWED UP CLOSE Seurat's massive painting, *La Grande Jatte*, reveals his careful decision-making about the hue and tone of every dot. But viewed at a specific distance, the spatial resolution of the pointillism begins to match the spatial resolution of color vision. Just when cone cells start mixing the information from neighboring dots, the painting starts to shimmer. Pointillism is one step towards a new kind of art that directly strives for psychological sensation in the viewer. Seurat worked hard on this painting, Twenty-three preliminary drawings and thirty-eight painted color studies survive to testify to his struggle toward a desired effect.

Figure 35: The juncture between dress, grass, and shadow reveal a variety of color choices, no single tone or hue.

Figure 36: *A Sunday on La Grande Jatte* George Seurat, 1884-1886. His best known painting is also his largest, 81.7 in x 121 x in. Art Institute of Chicago.

We do not see color using individual pixels. Information from individual cone cells is not sent directly to the brain. Instead, the retina integrates over areas of color information, comparing and contrasting the information from adjacent fields to decide what color to assign to each field. The brain uses color differences at boundaries to assign color, analogous to how it uses sudden differences in brightness to detect edges. Color depends on context.

Josef Albers, 1888-1976, was a German artist and educator who wrote the seminal work on color theory of artists, *The Interaction of Color.* Here, I will select a few of his lessons.

First, the same color has many faces. Color is relative.

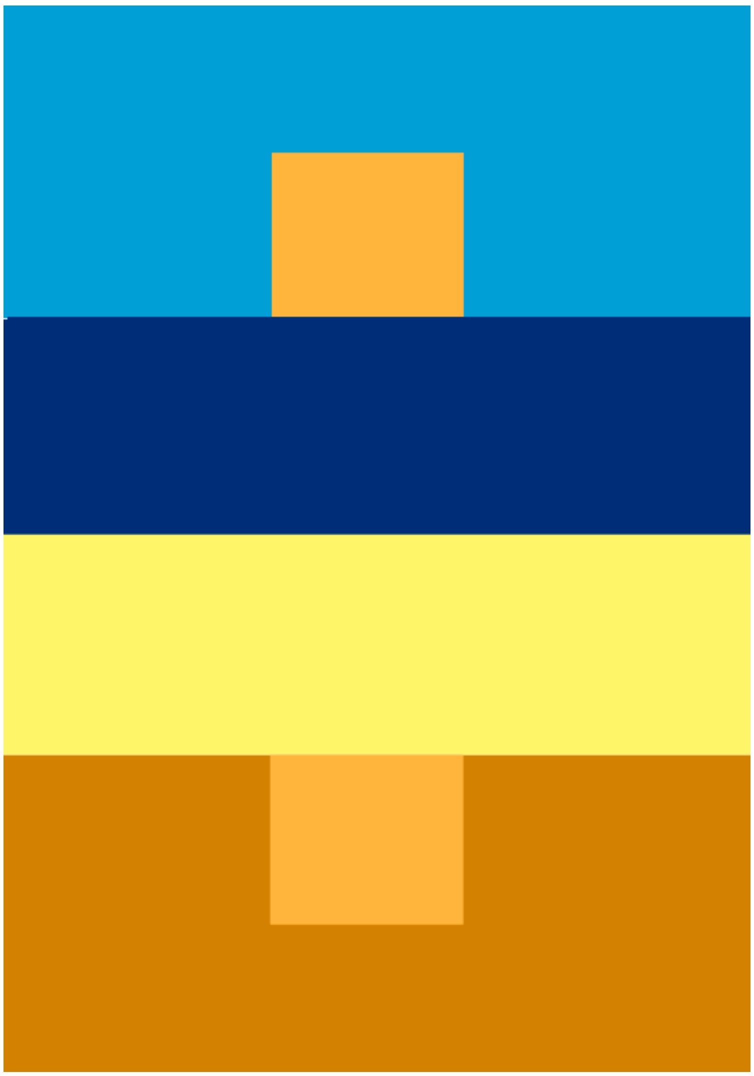

Figure 37: Plate IV-1. The upper small and the lower small squares are the same color, although the upper one looks more 'orange' and the lower one looks more 'tan'.

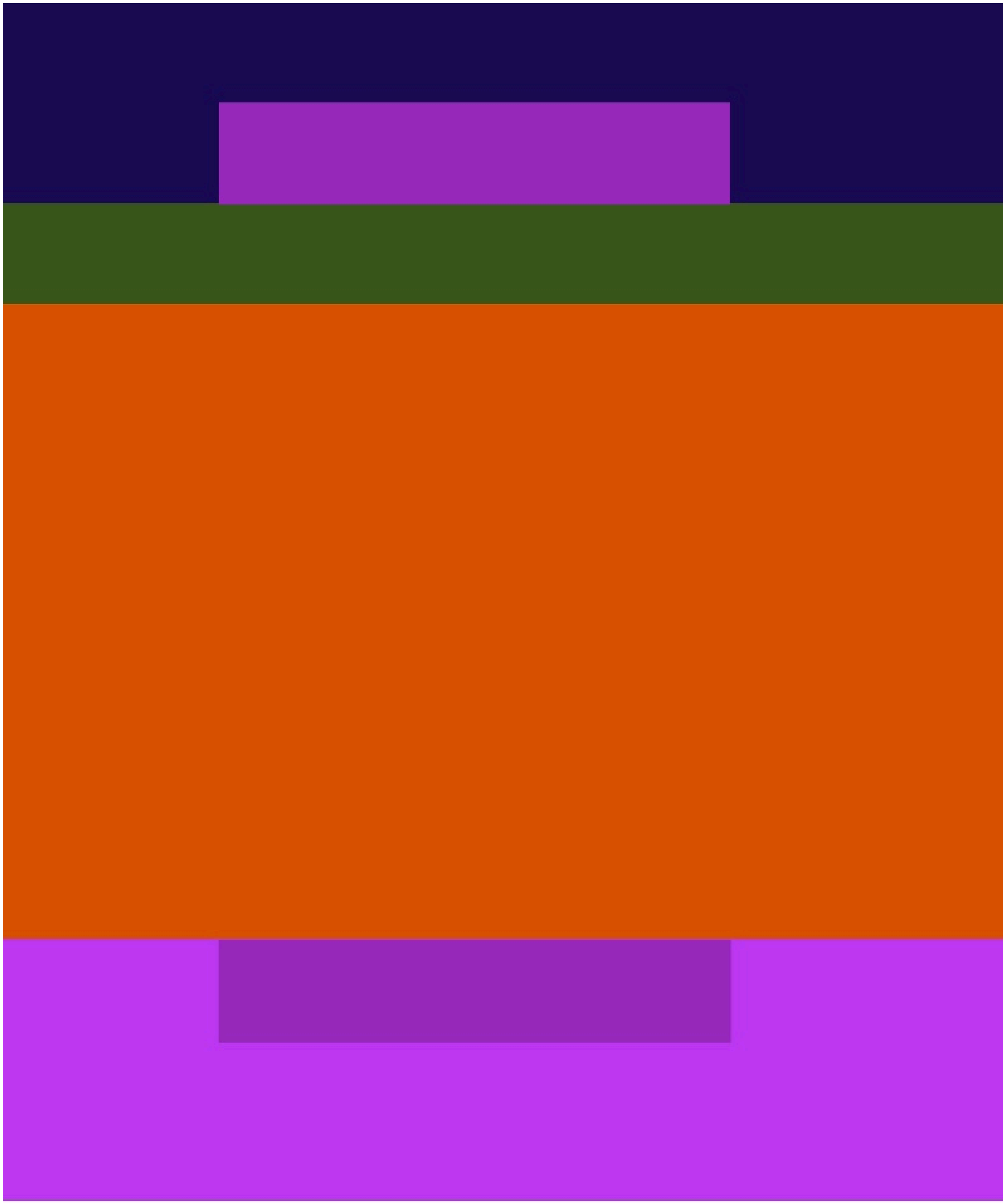

Figure 38: Plate IV-4. The inner smaller violets are the same, but the upper one looks more like the outer violet at the bottom.

AFTER-IMAGES are another way to experience how the eye determines color. Stare at the red circle for 30 s, keeping your eye on the black center. Then look at the black center of the white circle. It does not appear white. While you stare at the red circle, the retina is adapting. The photoreceptors for red light are strongly activated, and gradually begin to inhibit the output of neighboring red photoreceptors. This inhibition is long-lasting. So when you look at the white circle, the retina is effectively subtracting the red light from the white light, leaving blue/green light.

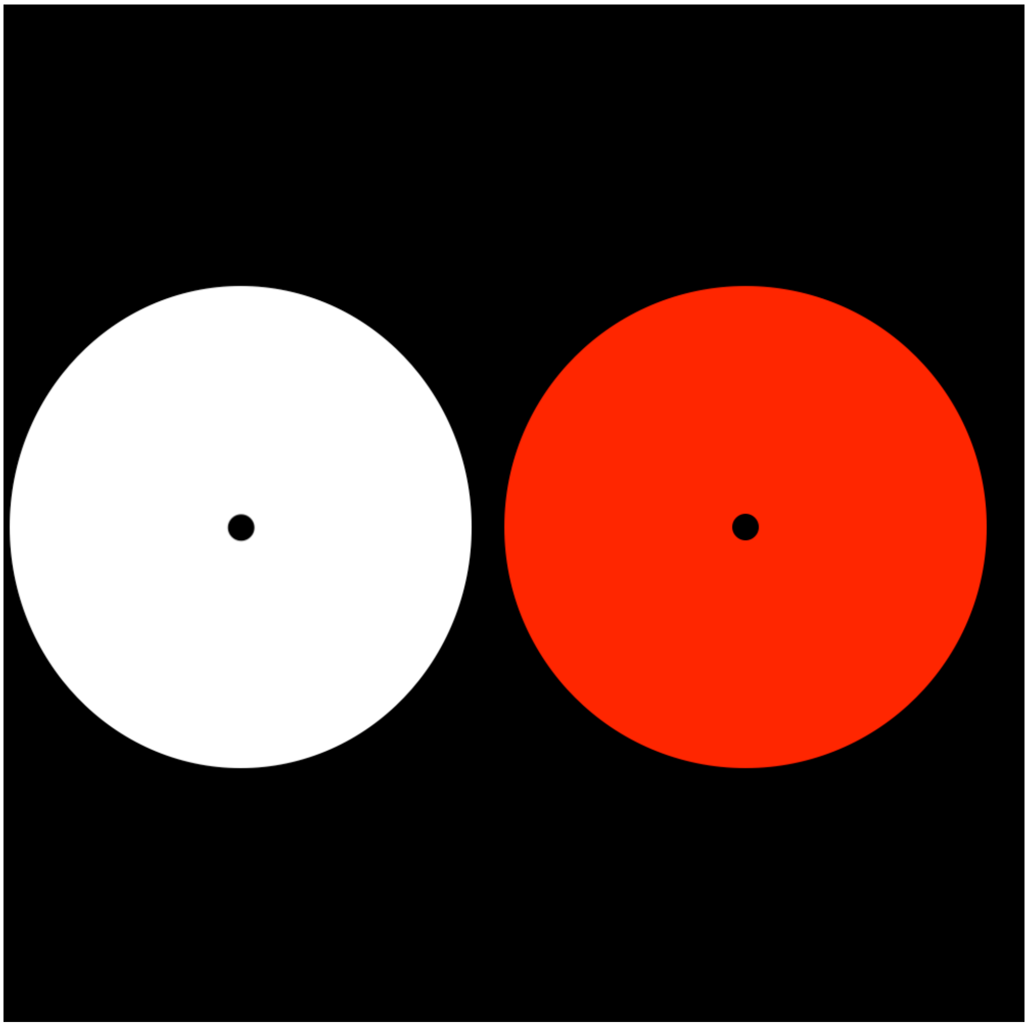

Figure 39: Plate VIII-I.

Double-reversal is seen in this experiment. After staring at the set of yellow circles in the white box, the eye gradually adapts by inhibiting yellow perception inside the circles. When you shift your eye to the white box, yellow diamonds appear.

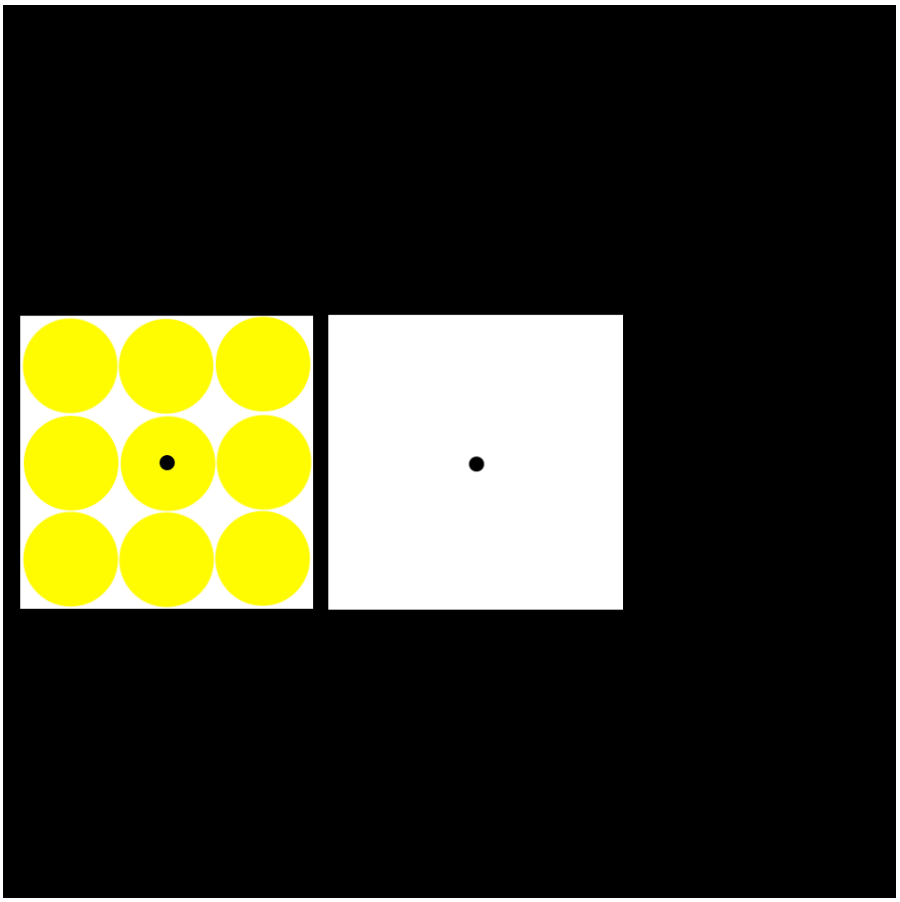

Figure 40: Plate VIII-2.

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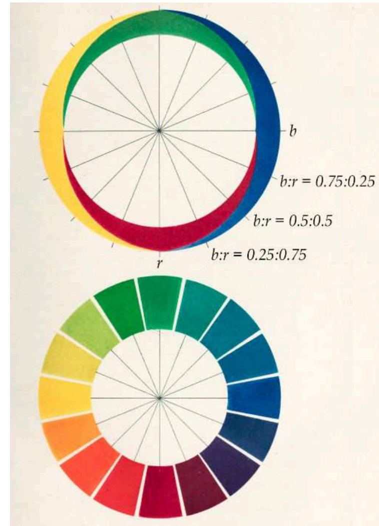

Figure 41: Hering's Color Opponency Theory, from Conway et al. (2023).

THE COLOR-OPPONENCY THEORY came from the systematic analysis of 'after-image' effects using different colors, and from hue-cancellation effects where one is asked what colored light will cancel a test color. Color-opponency theory is the basis of the color wheel that is widely used by artists.

While color-coding in the retina might be relatively straightforward – based on three cone types and inhibitory and cross-inhibitory circuitry – there is no simple algebra that converts retinal information into a reliable prediction of how each person sees color in any picture. Each human brain takes the whole field-of-view into account when deciding colors. Color-opponency theory – derived from simple experiments and color arrangements – does not fully explain color perception in art of life. A viral meme from 2015 (#thedress) makes this point.

CLAUDE MONET (1840-1926) uses color-opponency when he uses blue to paint the shadows of haystacks in yellowing twilight.

Figure 43: *Wheatstacks, Snow Effect, Morning* by Claude Monet, 1891. Getty Museum.

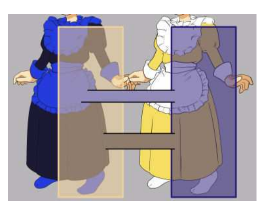

Figure 42: When this photo of a dress was posted to Facebook in 2015, it went viral. Many saw a blue and black dress. Many saw a white and gold dress. The answer depends on how you interpret the illuminating light, whether you see a blue and black dress that is illuminated by yellowish light or a white and gold dress that is illuminated by bluish light.

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Color vision assigns the hue of an object, but not identify the edges of an object. We use brightness information – spatial differences in luminosity – to find object boundaries. Josef Albers makes this clear with overlapping bars of different color and brightness. The lower horizontal bar appears to overlap the center bar, whereas the upper horizontal bar is overlapped by the center bar. The middle bar is ambiguous. The brain draws outlines around adjacent visual fields, connecting fields into single objects when similar contrasts in brightness occur at a shared outer edge.

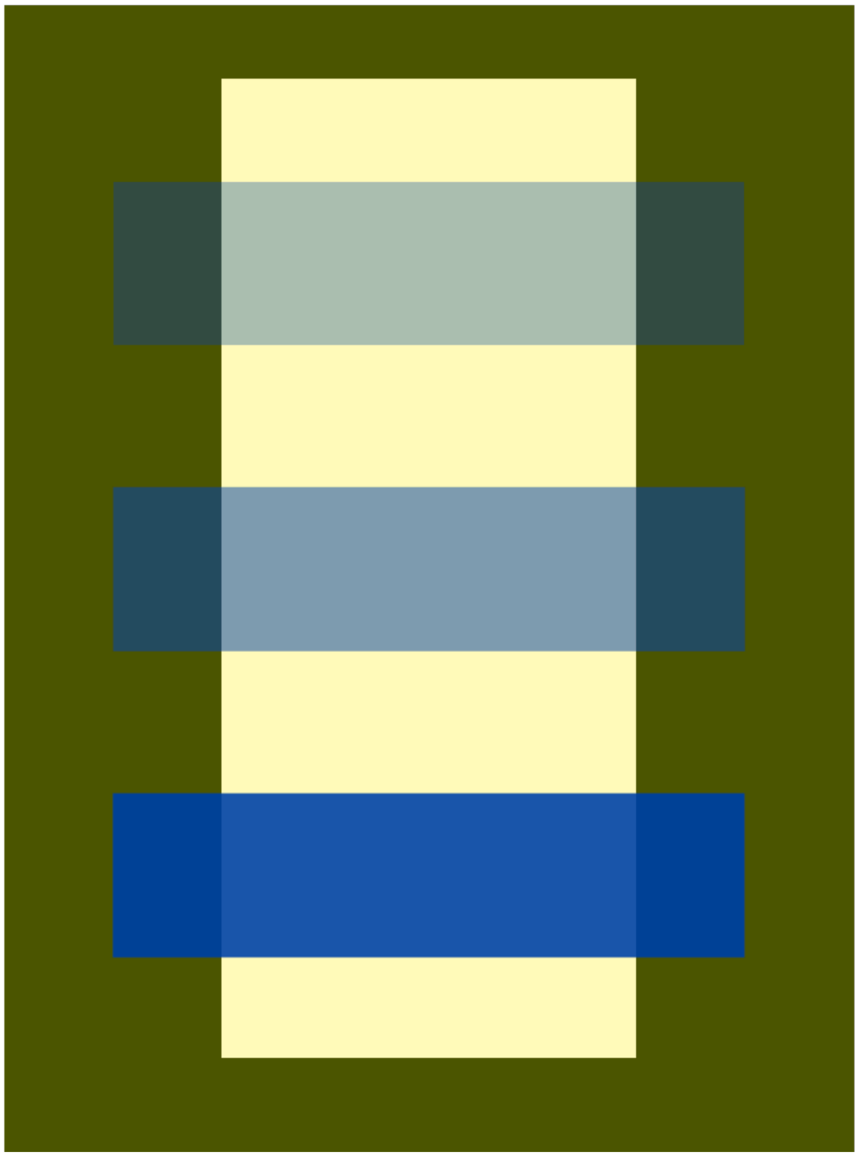


Figure 44: Plate XI-3.

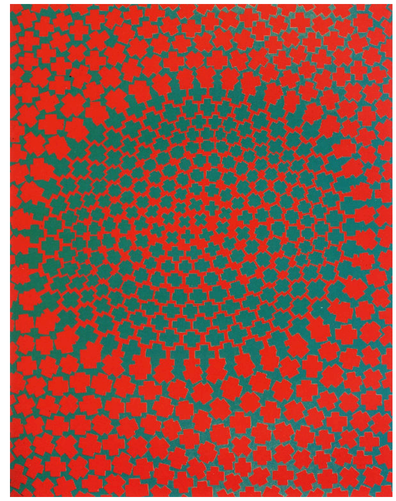


Figure 45: *Plus Reversed Sunrise* by Richard Anuszkiewciz, 1960. The equally luminous green and red fields shimmer and dance.

IMPRESSION, SUNRISE was painted by Monet in 1872. The name of the impressionist movement came from this painting of the port of Le Havre. The haziness and loose brush strokes broke from traditional landscape painting. Objects are hardly distinguishable and boundaries seem to vanish. Edge detection based on sudden changes in brightness is subverted. The sun shimmers. The sun lacks a well-defined boundary in the cloudy sky.

HOW DOES MONET CREATE THE ILLUSION of the shimmering sun? He used color mixes for sun and sky with equal brightness. Because the brain uses brightness, not color, to draw object outlines, it has trouble finding the edges of the sun. And so the sun appears to pulse and vibrate in the sky. This is made clear by shifting the painting to grayscale – the sun disappears.

PLUS REVERSED, oil on canvas, by Richard Anuszkiewciz (1930-2020) is a conscious effort toward pulsation and vibration by using isoluminous regions of red and green. The effect can be unsettling and is usually avoided by graphic designers. But an artist might want to unsettle their viewer.

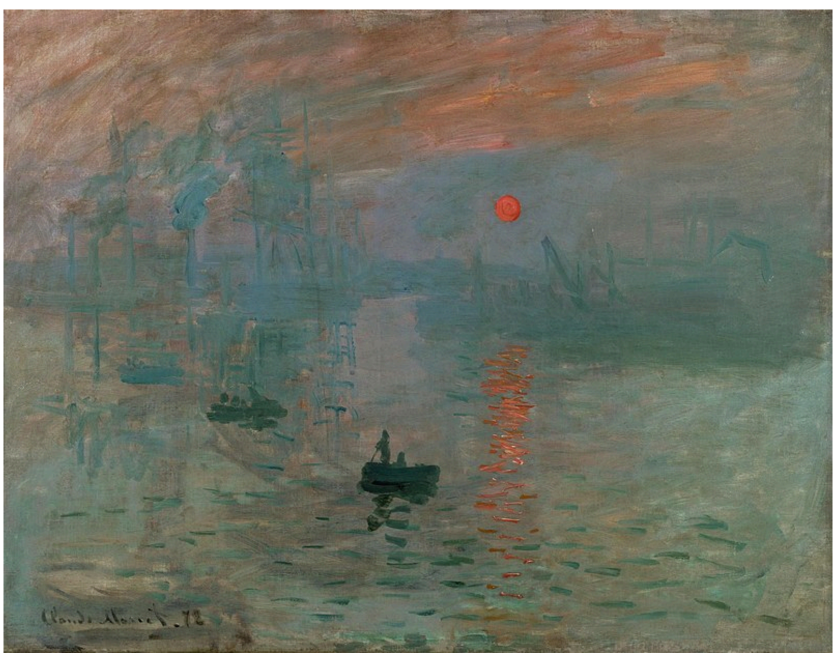
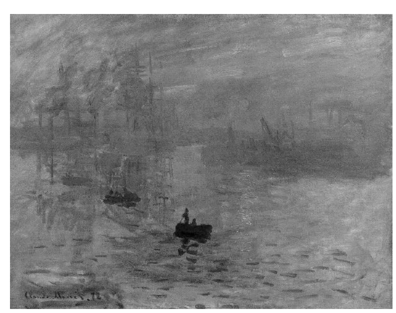


Figure 46: *Impression, Sunrise* by Claude Monet. Below, the same painting in grayscale. Almost all luminosity boundaries disappear, including between sun and the sky. By subverting your brain's ability to see object boundaries using luminance, the brain is confused.



Ochre and pink are easily recognized as different colors when presented separately. But when equal in luminosity, it becomes difficult to see the boundary between these colors when they are adjacent.

Figure 47: Plate XXIII-1

ANDRÉ DERAIN uses a range of different colors when painting the face of Henri Matisse. But by using equal luminosity in the many-colored brush strokes, the face keeps its coherence and is unfragmented at color boundaries.

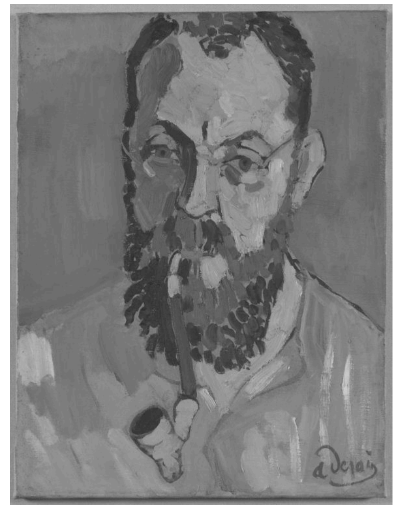


Figure 48: *Portrait of Henry Matisse* by André Derain, 1905. Tate Gallery.

IMPRESSIONISM broke from Western tradition in many ways. From the Renaissance to the 19th century, art was largely narrative. Pictures told stories or faithfully transcribed what was seen. Color was largely decorative, essentially filling in what could have been (or was) drawn.

The impressionists used color to compose the image. *Impression, Sunrise* and *La Grande Jatte* are painted with color and light. Without color, their images dissolve. In this way, Monet and Seurat released visual art from prior assumptions.

PAUL CEZANNE (1839-1906) bridged impressionism and modern art by finding new ways to release art from prior assumptions. The Dutch still lifes of Pieter Claesz or Willem Kalf were displays of virtuosity and optical precision. Cezanne painted many still lifes, where depth is modeled with color, not with linear perspective or shading. Nature is not intentionally distorted, but distortion was acceptable if the still life was no less convincing.

Figure 49: *Still Life with Commode* by Paul Cezanne, 1887-1888. Harvard Art Museums.

MODERN ART has made total break from narrative purpose. Mark Rothko (1903-1970) essentially painted nothing but color and light. A 'Rothko' means a canvas filled with floating blocks of color. Painting pure color, Rothko was strict about the lighting and placement of his canvases.

Figure 50: *No. 3/No. 13* by Mark Rothko, 1949. MoMA.

In 1962, Rothko painted ten murals for a dining room on the top floor of Holyoke Center at Harvard. The murals were displayed until 1970s when they had to be removed because of damage and fading. Color photographs reveal what they originally looked like. In 2014, these paintings were brought out of storage and 'digitally restored' for a special exhibition at the Harvard Art Museums, using a projector to precisely compensate for their fading.

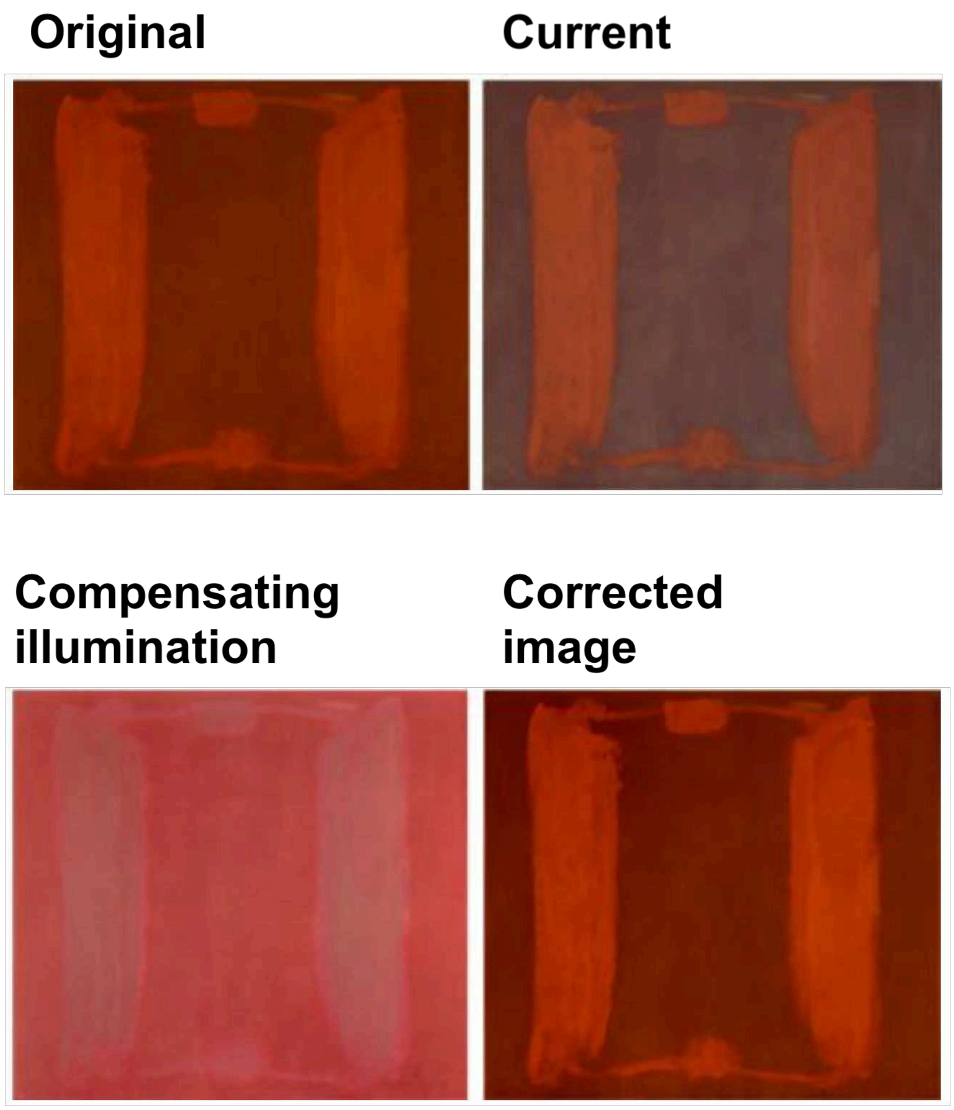


Figure 51: *Panel One* by Mark Rothko, 1962 and digital restoration.

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Updated: February 19, 2026

Shadow and Silhouette

Thursday, 12 February 2026, 12:45 - 2:15 PM EST.
Harvard Art Museums 0600

Silhouettes and shadow drawings stand at the beginning of Western Art. The Chauvet-Pont-d'Arc Cave in southeastern France contains some of the oldest known paintings in Europe. Made approximately 36,000 years ago, these charcoal drawings depict horses, rhinoceroses, and other hoofed animals. The artists selected the orientation that most clearly distinguishes each animal, providing just enough information to infer identity. These paintings were created deep within the cave, more than 300 meters from the surface, in spaces of absolute darkness except for torchlight. So they are unlikely to be simply decorative.

These early artists used profile views to represent hoofed mammals, intuitively selecting the most informative viewpoint for minimalist representation of the animal's body plan. In the same way, the human body is most intuitively drawn from the front, and a snake from above. The mind naturally seeks compact and informative representations of visual information. Picasso's *Le Tareau* (The Bull) is a modern study of this process – a lithographic sequence that strips away representational complexity towards its simplest and most essential structure.

Figure 52: Panel of the horses. Chauvet Pont d'Arc Cave. Link to website

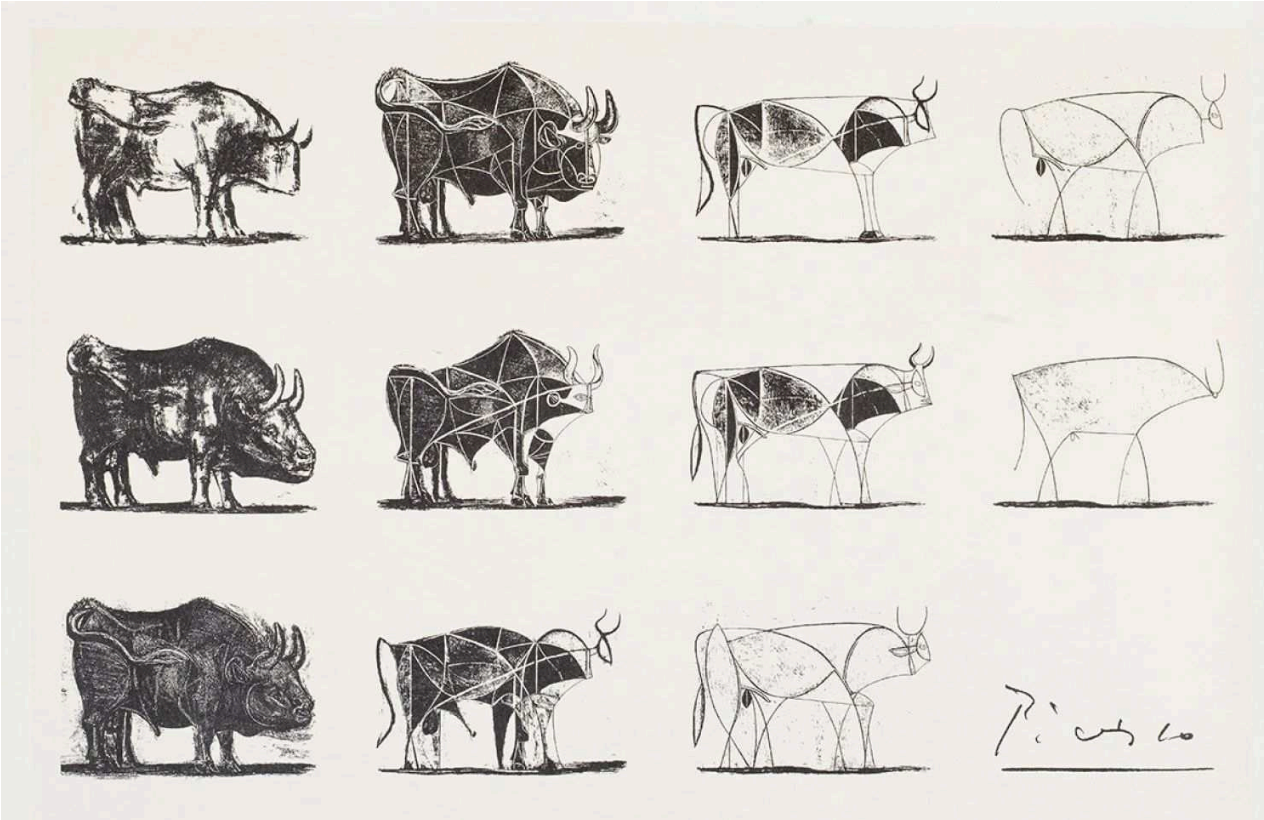

Figure 53: *Le Taureau* by Picasso, 1945-1946 Link to Moma

ACCORDING TO GREEK HISTORY, shadows played a role in the beginnings of art. Pliny the Elder (AD 23/24 - 79), in his *Natural History*, recounts an origin story about portraiture. Butades, a potter in Ancient Corinth, noticed that his daughter had traced the shadow of her partner cast by lamplight. Butades filled the outline with clay, thereby inventing the first portrait sculpture.

This story became a popular subject in European painting during the 18th and 19th centuries, when interest in Greek and Roman antiquity was revived. The cast shadow is the simplest interplay of light that can be exploited to create realistic representation. Of the human head, only the side profile allows different individuals to be easily distinguished. In antiquity, profile portraits were most widely circulated and seen on coins. Profiles on coins are not far removed from simple silhouettes. Because nobility and authority were so often represented in profile, the profile itself acquired an ennobling status.

Figure 54: Alexander the Great, 336-323 BCE. Harvard Art Museums

Figure 55: Caesar Augustus, Silver coin, 19-18BC. British Museum

Figure 56: Head of Brutus, Roman Republic, 43 -42 BC. Commemorative coin for Julius Caesar's assassination. British Museum

Figure 57: Johann and Heinrick Olivier, *Invention of the Art of Drawing*, 1805. Link to Metropolitan Museum of Art.

THE FLORENTINE RENAISSANCE witnessed an increase in illusionism, including the development of linear perspective — a mathematical system for projecting three-dimensional scenes onto two-dimensional surfaces. Modeling shadows and cast shadows also became tools for creating the illusion of volume. In *Trattoto della Pittura*, notes on painting, Leonardo da Vinci (1452-1519) strongly advocated the study and use of shadow.

> The art of painting consists in this, that by means of light and shade it makes bodies appear in relief upon a flat surface. Shadow is the means by which bodies display their form. The forms of bodies could not be understood in detail but for shadow.
>
> The painter who neglects light and shade deprives his figures of that relief which is the soul of painting; for without them the figures will appear flat and without life.

Leonardo trained in the workshop of Andrea del Verrocchio (1435-1488), who insisted that his apprentices master the depiction of light and shadow. In Verrocchio's workshop, Leonardo honed his skill through drapery studies — careful renderings of cloth arranged to reveal how light falls across folds. Leonardo continued drapery studies throughout his career.

Figure 58: *Drapery studies*, Leonard da Vinci, 1473-1477 Link to Louvre

Figure 59: Drapery study for *The Virgin and Child with Saint Anne* by Leonardo da Vinci, 1507-1510 Link to Louvre

In the Florentine Renaissance, most painting was commissioned by wealthy patrons. Painters were told *what* to paint, but chose *how* to paint. In 1472, Verrocchio received a commission for an *Annunciation* and gave it to Leonardo, his best apprentice.

In most paintings of the *Annunciation,* Gabriel and Mary are indoors or in enclosed spaces. Breaking from tradition, Leonardo chose an outdoor setting, allowing him to explore natural sunlight and its effects. The sun shines from the left, so both Gabriel and Mary cast shadows to the right. The direction of these shadows seems to literalize the Gospel of Luke's phrase: "The Holy Spirit will come on you, and the power of the Most High will *overshadow* you."

Another innovation was using oil paint, which had recently arrived in Florence from Northern Europe. Verrocchio painted in tempera, fast-drying pigment mixed with egg. Slow-drying oil paint allows layered refinement and revision. Technical imaging reveals that Gabriel's profile was originally lower and later raised so that he looks more directly at Mary. The Virgin's head was originally more upright, then reworked downward and slightly turned, creating a humbler gesture.

In most of the panel, Leonardo carefully transferred preparatory drawings by pouncing — pricking small holes along drawn lines and dusting charcoal dust to transfer the design. Much of the *Annunciation* shows evidence of pouncing, though the drapery appears to have been painted freehand, drawing on Leonardo's practiced mastery.

Figure 60: The Annunciation by Leonardo da Vinci, 1507-1510 Link to Uffizzi